ModelIt!

Student Activity Pack

Why Does Your Brain Lie to You?

Modeling Cognitive Bias, Perception, and Decision-Making in the Brain

Grade 12L1 | NGSS HS-LS1-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever been completely sure you saw or heard something that turned out to be different from what actually happened? Describe what occurred.**

2. **Why do you think two people can look at the same evidence and reach completely opposite conclusions?**

3. **What do you think a 'cognitive bias' is, and can you name any examples?**

4. **Do you think smarter people are less biased? Explain your reasoning.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sensory Input | Prior Experience | Cognitive Load | Bias Strength | Decision Accuracy

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Clear Evidence

Sensory Input: High | Prior Experience: Low | Cognitive Load: Low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Strong Priors

Sensory Input: Medium | Prior Experience: Maximum | Cognitive Load: Low

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Overloaded Brain

Sensory Input: Low | Prior Experience: High | Cognitive Load: Maximum

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Emotional State — The current emotional condition of the decision-maker — fear, anger, and excitement each amplify different biases and override rational processing in predictable ways

Social Pressure — Influence from peers, authority figures, or group consensus — conformity bias can override personal judgment even when the individual knows the group is wrong

Metacognition — Awareness of one's own thinking processes — acts as a moderating variable that can reduce but never eliminate the impact of cognitive biases on decisions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If your brain is the most powerful computer on Earth, why does it constantly trick you into seeing things that aren't there and believing things that aren't true? Use evidence from your simulation to support your answers.

1. **How does your model explain why two scientists can look at the same data and reach different conclusions?**

2. **Why does increasing cognitive load make biases stronger, and what does this imply for important decisions made under stress?**

3. **If cognitive biases are built into how the brain works, can we ever truly eliminate them? What does your model suggest?**

4. **How might understanding cognitive bias change the way you evaluate news, social media, or advertising?**

5. **What are the limitations of a five-component model for something as complex as human decision-making?**

8 STEM Challenge: Design a Bias-Aware Decision Tool

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YOUR ENGINEERING MISSION

Design a decision-making tool or protocol that helps people recognize and counteract cognitive biases when making important choices about health, finances, or education.

THE SCENARIO

A hospital has noticed that doctors' diagnostic accuracy drops significantly during long shifts. Research shows cognitive biases like anchoring (fixating on the first diagnosis) and confirmation bias (only looking for evidence that supports the initial guess) are major contributors. Your team must design a decision-support tool that helps doctors maintain diagnostic accuracy even when cognitive load is high.

Engineering Design Thinking

1. WHAT SPECIFIC BIAS-INTERRUPTION MECHANISMS DOES YOUR TOOL INCLUDE AND HOW DOES EACH ONE WORK?
2. HOW DOES YOUR DESIGN ACCOUNT FOR THE FACT THAT DOCTORS UNDER HIGH COGNITIVE LOAD RESIST COMPLEX PROTOCOLS?
3. WHAT EVIDENCE FROM YOUR MODEL SUPPORTS YOUR DESIGN CHOICES?
4. HOW WOULD YOU TEST WHETHER YOUR TOOL ACTUALLY REDUCES DIAGNOSTIC ERRORS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Cognitive scientists and behavioral researchers study how the brain processes information and makes decisions. They work in healthcare, tech companies, government policy, and academia, earning

$65,000-$130,000/year. UX researchers who apply cognitive science to product design earn $80,000-$150,000/year.